

# An Eight-Charge Counterexample to the Maxwell Bound

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## Abstract

We construct a configuration of eight distinct positive point charges in three-dimensional Euclidean space whose Coulomb potential has a finite critical set of cardinality at least 51, with every critical point nondegenerate. Since  $51 > (8 - 1)^2 = 49$ , this gives an eight-charge counterexample to the Maxwell bound. Our construction begins with a five-charge configuration having at least 24 nondegenerate critical points and adds three positive axial charges on two separated local scales. A polynomial limit accounts for 17 persistent critical points, while an innermost point-charge limit contributes 11 more. Together with the 23 nonzero critical points inherited from the five-charge seed, the total is 51.

## 1 Introduction

Let  $q_1, \dots, q_n > 0$ , and let  $a_1, \dots, a_n \in \mathbb{R}^3$  be pairwise distinct. The associated Coulomb potential is

$$V(x) = \sum_{i=1}^n \frac{q_i}{|x - a_i|}$$

on  $\mathbb{R}^3 \setminus \{a_1, \dots, a_n\}$ . The equilibria of the electric field  $-\nabla V$  are precisely the critical points of  $V$ . A critical point is *nondegenerate* if its Hessian is invertible, and the charge configuration is *Morse* if all critical points of  $V$  are nondegenerate.

In the positive-charge setting considered here, the form of Maxwell’s conjecture relevant to this paper is the following.

**Maxwell’s conjecture.** For every Morse configuration of  $n$  distinct positive point charges in  $\mathbb{R}^3$ , the number  $N$  of critical points satisfies

$$N \leq (n - 1)^2. \tag{1}$$

Historical formulations differ in whether signed strengths are allowed and in the genericity hypotheses imposed on the equilibria. Throughout this paper, “Maxwell’s conjecture” means precisely the positive-charge Morse statement above. Although a Morse function on a noncompact domain need not have finitely many critical points, finiteness follows here from positivity: every critical point lies in the convex hull of the charge positions, and a neighborhood of each charge is free of critical points. The critical set of a Morse potential is therefore a compact discrete set and hence is finite. We give the details in Section 8.

Maxwell discussed points of equilibrium in his 1873 treatise [1]. Morse and Cairns later posed the problem of finding a general upper bound [2]. Gabriellov, Novikov, and Shapiro obtained the

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first general finite bound in 2007 and, in particular, the upper bound 12 for three charges [3]. Tsai proved the conjectured upper bound of four for three equal positive charges [4], and Lee–Tsai exhibited a positive coplanar four-charge configuration with nine equilibria [5]. Zolotov subsequently improved the general bound using the Thom–Milnor theorem [6]. Edelsbrunner, Fillmore, and Oliveira proved the bound  $2^n(3n-2)^3$  for a generic choice of one charge position and constructed families whose equilibrium-to-charge ratio tends to 25/7 [7]. In 2026, Arathoon–Ball–Kvalheim gave the first counterexample to Maxwell’s conjecture: five positive charges with at least 24 nondegenerate critical points [8].

We next describe precisely how the five- and seven-charge results arise and why the same count does not by itself settle the eight-charge case. Write  $e_z = (0, 0, 1)$ . The five-charge construction of Arathoon–Ball–Kvalheim starts with three unit charges at the vertices of an equilateral triangle centered at the origin in the plane  $z = 0$  and adds equal positive charges

$$q_\epsilon = \frac{3}{4}\epsilon^3 - \frac{5}{32}\epsilon^5$$

at  $\pm\epsilon e_z$ . On the scale  $x = \epsilon^2 X$ , the central critical point gives rise to a local family of 21 nondegenerate critical points, including the origin. The three noncentral critical points of the triangle persist and remain separated from this local family. Thus the five-charge potential has  $N_5 \geq 21 + 3 = 24 > 16$ .

Their iteration is recorded in Remark 3 and Proposition 1 of [8]. Suppose a  $D_3 \times \mathbb{Z}_2$ -symmetric potential has the expansion

$$U(x) = U(0) + \alpha H_2(x) + \beta H_3(x) + O(|x|^4), \quad \alpha, \beta > 0,$$

where  $D_3$  acts in the  $(x, y)$ -plane,  $\mathbb{Z}_2$  acts by  $z \mapsto -z$ , and  $H_2, H_3, H_4$  are defined in Section 2. Add equal charges of strength  $q_\eta^{\text{iter}} = \alpha\eta^3 - \hat{\gamma}\eta^5$  at  $\pm\eta e_z$ ; this strength is positive for all sufficiently small  $\eta > 0$ . With

$$\lambda = \frac{2\beta}{5\alpha}, \quad \hat{\gamma} = \frac{\beta^2}{30\alpha},$$

the rescaling  $x = \lambda\eta^2 X$ , subtraction of the value at the origin, and division by  $\eta^6$  produce the polynomial limit

$$\frac{64\beta^4}{1875\alpha^3} \left( \frac{5}{32}H_2 + \frac{15}{8}H_3 + \frac{3}{16}H_4 \right).$$

The prefactor is positive, so this limit has the same 21 nondegenerate critical points as the polynomial in the five-charge construction. One of the 21 points is the origin, which was already a critical point before the new pair was added; the other 20 give the certified net gain. All previously existing nonzero nondegenerate critical points persist. Starting from the equilateral triangle and perturbing only after all  $m$  iterations have been completed, Proposition 1 of [8] therefore gives, for every integer  $m \geq 0$ , a positive Morse configuration satisfying

$$n = 3 + 2m, \quad N \geq 4 + 20m, \tag{2}$$

with a finite critical set.

For  $m = 1$ , equation (2) is the five-charge lower bound  $N_5 \geq 24$ . For  $m = 2$ , it gives the seven-charge counterexample  $N_7 \geq 44 > 36 = (7-1)^2$ . It is important that this lower bound does not certify further violations. Its excess over Maxwell’s quantity is  $(4+20m) - (2m+2)^2 = 4m(3-m)$ , which is strictly positive only for  $m = 1, 2$ . The lower bound equals Maxwell’s quantity for  $m = 0$  and  $m = 3$ , corresponding to  $n = 3$  and  $n = 9$ , and is smaller for  $m \geq 4$ . This comparison concerns only the certified lower bound; it is not an upper-bound statement about the iterated configurations.

Equivalently, for every odd  $n \geq 3$ , Proposition 1 of [8] supplies a positive Morse configuration with  $N \geq 10n - 26$ . For every even  $n \geq 4$ , start from the corresponding Morse  $(n - 1)$ -charge configuration and place one additional positive charge sufficiently far away. Choose its position outside the plane of the three triangle charges when  $n = 4$ ; for  $n \geq 6$ , the existing positions are already noncoplanar. On fixed neighborhoods of the finitely many old critical points, the new charge has arbitrarily small  $C^2$ -norm when its position is sufficiently distant. Hence every old critical point persists and retains its Morse index. The strength-transversality argument of Section 8 then permits an arbitrarily small perturbation of the positive strengths that makes the full configuration Morse without destroying these persistent points. In the notation of Proposition 11, the persistent points have  $m_2 - m_1 = n - 2$ , whereas the full  $n$ -charge configuration must have  $m_2 - m_1 = n - 1$ . At least one additional critical point must therefore exist, and this one-charge extension gives

$$N \geq 10n - 35 \quad (n \geq 4 \text{ even}).$$

For the latter lower bound,  $(10n - 35) - (n - 1)^2 = -(n - 6)^2$ , so this even-charge extension never proves a strict violation and reaches equality only at  $n = 6$ . Thus, among the bounds furnished by the iteration and this one-charge extension, strict violations occur only for  $n = 5$  and  $n = 7$ . The conjecture is elementary for  $n = 1, 2$ . At the time of writing, it remained unresolved in full generality for  $n = 3, 4, 6, 8$ , and the preceding constructions do not decide it for any  $n \geq 9$ . Our Theorem 1 resolves the  $n = 8$  case negatively.

The seven-charge stage is rederived below in notation adapted to the present construction. After fixing the five-charge seed  $W$ , whose expansion has positive coefficients  $A, B$ , we add a second equal pair of strength  $A\eta^3 - \gamma\eta^5$  at  $\pm\eta e_z$ . A polynomial limit has exactly 21 nondegenerate critical points, while 23 selected nonzero critical points of  $W$  persist, giving  $23 + 21 = 44$ . The eighth charge is one-sided, axial, and much smaller. It produces 11 nondegenerate critical points on a still smaller scale, while a quantitative persistence argument preserves the 17 outer points of the 21-point polynomial pattern. Scale separation makes these two local families mutually disjoint and separates them from the 23 seed points. Thus the certified count is  $23 + (17 + 11) = 51$ . We neither need nor claim that the other four points of the polynomial pattern disappear after the eighth charge is added.

**Theorem 1** (Eight-charge counterexample). *There is a configuration of eight distinct positive point charges in  $\mathbb{R}^3$  whose Coulomb potential is Morse with a finite critical set of cardinality  $N \geq 51$ . Since  $51 > (8 - 1)^2 = 49$ , the Maxwell bound fails for  $n = 8$ .*

We use the following five-charge result of Arathoon–Ball–Kvalheim [8].

**Theorem 2** (Five-charge seed). *There is an  $\epsilon_0 > 0$  such that, for every  $0 < \epsilon < \epsilon_0$ , one may place three unit charges at the vertices of an equilateral triangle of circumradius one, centered at the origin in the plane  $z = 0$ , and equal positive charges*

$$q_\epsilon = \frac{3}{4}\epsilon^3 - \frac{5}{32}\epsilon^5$$

*at  $\pm\epsilon e_z$ . The resulting Coulomb potential has at least 24 distinct nondegenerate critical points. It may have additional critical points, possibly degenerate ones.*

At most one of the 24 distinct points in Theorem 2 can be the origin, so at least 23 lie away from it. In Section 2 we also verify directly that the origin is itself a nondegenerate critical point of the five-charge potential. We fix any 23 of the nondegenerate critical points away from the origin for the persistence argument below.

In the rest of this paper, we construct the three additional charges and prove Theorem 1. Section 2 establishes the local expansion of the imported seed. Sections 3–4 define the three new

charges and obtain the first local limit. Sections 5–7 prove the  $17 + 11$  local count and combine it with the 23 persistent seed points. Section 8 removes any remaining degeneracies and proves finiteness. Finally, Section 9 proves the index identity used above and checks the Morse-index ledger for all 51 constructed points.

## 2 Notation and the five-charge seed

Recall that  $e_z = (0, 0, 1)$ . In Cartesian coordinates  $(x, y, z)$ , put  $\rho^2 = x^2 + y^2$ ; when  $\rho > 0$ , let  $\theta$  denote the azimuthal angle, so that  $x = \rho \cos \theta$  and  $y = \rho \sin \theta$ . Define

$$H_2 = \rho^2 - 2z^2, \quad H_3 = x^3 - 3xy^2 = \rho^3 \cos(3\theta),$$

and

$$H_4 = 3\rho^4 - 24\rho^2 z^2 + 8z^4.$$

These are homogeneous polynomials of degrees 2, 3, and 4, respectively, and direct differentiation gives  $\Delta H_2 = \Delta H_3 = \Delta H_4 = 0$ . Thus all three are harmonic. The cylindrical expression for  $H_3$  is used only when  $\rho > 0$ ; its Cartesian expression defines it everywhere.

The triangle vertices are

$$a_1 = (1, 0, 0), \quad a_2 = (-1/2, \sqrt{3}/2, 0), \quad a_3 = (-1/2, -\sqrt{3}/2, 0).$$

They satisfy  $|a_j| = 1$  and  $|a_j - a_k| = \sqrt{3}$  for  $j \neq k$ . Thus the triangle is equilateral with circumradius one (and side length  $\sqrt{3}$ ).

Let  $\epsilon_0$  be as in Theorem 2, and fix  $0 < \epsilon < \min\{\epsilon_0, 1\}$ . Since

$$q_\epsilon = \epsilon^3 \left( \frac{3}{4} - \frac{5}{32} \epsilon^2 \right),$$

the quantity in parentheses is at least  $19/32$ , and hence the axial strength  $q_\epsilon$  is positive. Let  $W$  denote the five-charge potential

$$W(x) = \sum_{j=1}^3 \frac{1}{|x - a_j|} + \frac{q_\epsilon}{|x - \epsilon e_z|} + \frac{q_\epsilon}{|x + \epsilon e_z|},$$

defined on  $\mathbb{R}^3 \setminus \{a_1, a_2, a_3, \epsilon e_z, -\epsilon e_z\}$ .

**Proposition 3** (Expansion of the seed potential). *The potential  $W$  is real analytic near the origin and has the expansion*

$$W(x) = W(0) + AH_2(x) + BH_3(x) + O_\epsilon(|x|^4), \quad (3)$$

where

$$A = \frac{5}{32} \epsilon^2 > 0, \quad B = \frac{15}{8} > 0. \quad (4)$$

More precisely,

$$W(x) = W(0) + \frac{5}{32} \epsilon^2 H_2(x) + \frac{15}{8} H_3(x) + \left( \frac{3}{16\epsilon^2} + \frac{13}{128} \right) H_4(x) + O_\epsilon(|x|^5),$$

where  $W(0) = 3 + 2q_\epsilon/\epsilon$ . Define  $\mathcal{E}_\epsilon(x) = W(x) - W(0) - AH_2(x) - BH_3(x)$ . There are constants  $0 < r_\epsilon < \epsilon$  and  $C_\epsilon > 0$  such that, for every multi-index  $\alpha$  with  $|\alpha| \leq 2$ ,

$$|D^\alpha \mathcal{E}_\epsilon(x)| \leq C_\epsilon |x|^{4-|\alpha|} \quad (|x| < r_\epsilon).$$

*Proof.* Because  $0 < \epsilon < 1$ , the open ball  $B_\epsilon(0)$  contains no charge position, so  $W$  is real analytic there. We compute its Taylor polynomial through degree four.

Write  $V_\Delta(x) = \sum_{j=1}^3 |x - a_j|^{-1}$  for the potential of the three unit charges. For any unit vector  $a$ , Taylor expansion at the origin gives

$$\begin{aligned} \frac{1}{|x - a|} &= 1 + a \cdot x + \frac{1}{2}(3(a \cdot x)^2 - |x|^2) \\ &\quad + \frac{1}{2}(5(a \cdot x)^3 - 3|x|^2(a \cdot x)) \\ &\quad + \frac{1}{8}(35(a \cdot x)^4 - 30|x|^2(a \cdot x)^2 + 3|x|^4) + O(|x|^5). \end{aligned}$$

For the three vectors  $a_j$ , direct calculation gives

$$\sum_{j=1}^3 a_j = 0, \quad \sum_{j=1}^3 (a_j \cdot x)^2 = \frac{3}{2}\rho^2, \quad \sum_{j=1}^3 (a_j \cdot x)^3 = \frac{3}{4}H_3(x),$$

and

$$\sum_{j=1}^3 (a_j \cdot x)^4 = \frac{9}{8}\rho^4.$$

Substitution into the preceding Taylor formula yields

$$V_\Delta(x) = 3 + \frac{3}{4}H_2(x) + \frac{15}{8}H_3(x) + \frac{9}{64}H_4(x) + O(|x|^5).$$

For  $|x| \leq \epsilon/2$ , the generating function for the Legendre polynomials gives the absolutely and uniformly convergent series

$$\frac{q_\epsilon}{|x - \epsilon e_z|} + \frac{q_\epsilon}{|x + \epsilon e_z|} = \frac{2q_\epsilon}{\epsilon} \sum_{m=0}^{\infty} \left( \frac{|x|}{\epsilon} \right)^{2m} P_{2m} \left( \frac{z}{|x|} \right),$$

where each product  $|x|^{2m} P_{2m}(z/|x|)$  is understood as its homogeneous polynomial extension at  $x = 0$ . Using the formulas for  $P_2$  and  $P_4$  gives

$$\frac{q_\epsilon}{|x - \epsilon e_z|} + \frac{q_\epsilon}{|x + \epsilon e_z|} = \frac{2q_\epsilon}{\epsilon} - \frac{q_\epsilon}{\epsilon^3}H_2(x) + \frac{q_\epsilon}{4\epsilon^5}H_4(x) + O\left(\frac{q_\epsilon|x|^6}{\epsilon^7}\right).$$

In particular, the axial pair contributes no odd-degree terms. Using the definition of  $q_\epsilon$ , the combined quadratic coefficient is

$$\frac{3}{4} - \frac{q_\epsilon}{\epsilon^3} = \frac{3}{4} - \left( \frac{3}{4} - \frac{5}{32}\epsilon^2 \right) = \frac{5}{32}\epsilon^2,$$

whereas the cubic coefficient remains  $15/8$ . The combined quartic coefficient is

$$\frac{9}{64} + \frac{q_\epsilon}{4\epsilon^5} = \frac{9}{64} + \frac{3}{16\epsilon^2} - \frac{5}{128} = \frac{3}{16\epsilon^2} + \frac{13}{128}.$$

This proves both displayed expansions and the values in equation (4).

It remains to prove the derivative estimate. By the two expansions just obtained, the Taylor series of  $\mathcal{E}_\epsilon$  begins in degree four. Hence, for  $|\alpha| \leq 2$ , every derivative of  $D^\alpha \mathcal{E}_\epsilon$  of order strictly less than  $4 - |\alpha|$  vanishes at the origin. Choose  $0 < r_\epsilon < \epsilon/2$ . Taylor's theorem on  $\overline{B}_{r_\epsilon}(0)$ , applied to the finitely many derivatives with  $|\alpha| \leq 2$ , gives a common constant  $C_\epsilon$  for which

$$|D^\alpha \mathcal{E}_\epsilon(x)| \leq C_\epsilon |x|^{4-|\alpha|} \quad (|x| < r_\epsilon).$$

This completes the proof. □

The expansion has no linear term, so  $\nabla W(0) = 0$ . Its quadratic part is  $A(x^2 + y^2 - 2z^2)$ , and therefore  $D^2W(0) = \text{diag}(2A, 2A, -4A)$ . Since  $A > 0$ , the origin is a nondegenerate critical point of  $W$ .

Finally, let  $\mathcal{K} \subset \mathbb{R}^3$  be compact. For every multi-index  $\alpha$  with  $|\alpha| \leq 2$ , the chain rule gives

$$D_X^\alpha [\eta^{-6} \mathcal{E}_\epsilon(\eta^2 X)] = \eta^{2|\alpha|-6} (D^\alpha \mathcal{E}_\epsilon)(\eta^2 X).$$

For sufficiently small  $\eta > 0$ , one has  $\eta^2 \mathcal{K} \subset B_{r_\epsilon}(0)$ , and the derivative estimate above therefore yields, uniformly for  $X \in \mathcal{K}$ ,

$$|D_X^\alpha [\eta^{-6} \mathcal{E}_\epsilon(\eta^2 X)]| \leq C_\epsilon \eta^2 |X|^{4-|\alpha|}.$$

It follows that

$$\|\eta^{-6} \mathcal{E}_\epsilon(\eta^2 \cdot)\| * C^2(\mathcal{K}) = O * \epsilon, \mathcal{K}(\eta^2) \quad \text{as } \eta \downarrow 0.$$

In every subsequent limit,  $\epsilon$  is fixed first. Thus the constants implicit in  $O_\epsilon(\cdot)$  and in the derivative estimates above need not be uniform as  $\epsilon \downarrow 0$ .

### 3 The eight charges

Fix the dimensionless normalization constants

$$\kappa = 10, \quad \mu = \frac{1}{2}.$$

Let  $\gamma > 0$  be a parameter, to be fixed sufficiently small in Sections 5–7, and put

$$d_\gamma := \frac{\kappa\gamma}{B} = \frac{16}{3}\gamma, \tag{5}$$

where the second equality uses  $B = 15/8$ . The choice of  $d_\gamma$  gives  $Bd_\gamma/\gamma = \kappa$ . Indeed, since  $H_3$  is homogeneous of degree three, the change of variables  $X = d_\gamma Y$ , followed by division by  $\gamma d_\gamma^2$ , changes the coefficient of  $BH_3$  to  $Bd_\gamma/\gamma = \kappa$ . Under the preceding rescaling  $x = \eta^2 X$  and division by  $\eta^6$ , the eighth charge has coefficient  $Q_{\gamma,\eta}/\eta^8$ ; the same inner rescaling changes this coefficient to  $Q_{\gamma,\eta}/(\eta^8 \gamma d_\gamma^3)$ . We therefore choose that charge so that the latter ratio equals  $\mu$ . These are precisely the coefficients used in the innermost limit of Section 6.

For  $\eta > 0$ , add the following three charges to the five-charge configuration defining  $W$ :

1. Equal charges

$$p_{\gamma,\eta} = A\eta^3 - \gamma\eta^5 \tag{6}$$

at  $\pm\eta e_z$ .

2. One charge

$$Q_{\gamma,\eta} = \mu\gamma d_\gamma^3 \eta^8 = \frac{2048}{27} \gamma^4 \eta^8 \tag{7}$$

at  $\delta e_z$ , where

$$\delta = d_\gamma \eta^2 = \frac{16}{3} \gamma \eta^2. \tag{8}$$

Denote the resulting eight-charge potential by

$$V_{\gamma,\eta}(x) = W(x) + \frac{p_{\gamma,\eta}}{|x - \eta e_z|} + \frac{p_{\gamma,\eta}}{|x + \eta e_z|} + \frac{Q_{\gamma,\eta}}{|x - \delta e_z|}.$$

Here  $\epsilon$  has already been fixed in Section 2. The order of parameter selection is important: first  $\epsilon$  is fixed, then  $\gamma$  is fixed after all later smallness conditions on it have been imposed, and only then is  $\eta$  chosen. For fixed  $\gamma > 0$ ,

$$p_{\gamma,\eta} = \eta^3(A - \gamma\eta^2) > 0 \quad \text{if} \quad \eta^2 < \frac{A}{\gamma},$$

while  $Q_{\gamma,\eta} > 0$  automatically. In particular, every choice

$$0 < \eta < \min \left\{ \epsilon, \sqrt{\frac{A}{\gamma}}, \frac{1}{d_\gamma} \right\}$$

satisfies

$$p_{\gamma,\eta} > 0, \quad 0 < \delta = d_\gamma \eta^2 < \eta < \epsilon.$$

The five axial positions  $-\epsilon e_z, -\eta e_z, \delta e_z, \eta e_z, \epsilon e_z$  are therefore distinct, and none is a triangle vertex. Together with the three positive unit strengths and the positive seed strength  $q_\epsilon$ , this proves that the construction has eight distinct positions and eight positive strengths.

## 4 The first local limit

For an axial pair of strength  $p$  at  $\pm \eta e_z$ , write

$$U_{p,\eta}(x) = \frac{p}{|x - \eta e_z|} + \frac{p}{|x + \eta e_z|}.$$

Taylor expansion at the origin gives

$$U_{p,\eta}(x) = \frac{2p}{\eta} - \frac{p}{\eta^3} H_2(x) + \frac{p}{4\eta^5} H_4(x) + R_{p,\eta}(x). \quad (9)$$

There is an absolute constant  $C > 0$  such that, for every multi-index  $\alpha$  with  $|\alpha| \leq 2$ ,

$$|D^\alpha R_{p,\eta}(x)| \leq C|p|\eta^{-7}|x|^{6-|\alpha|} \quad (|x| \leq \eta/2). \quad (10)$$

Indeed, after writing  $x = \eta u$ , this is Taylor's theorem on  $\overline{B}_{1/2}(0)$  for the analytic even function  $u \mapsto |u - e_z|^{-1} + |u + e_z|^{-1}$ , followed by rescaling. Its terms of degrees zero, two, and four are precisely those displayed in equation (9); evenness makes the remainder begin in degree six.

Set

$$\tilde{V}_{\gamma,\eta}(X) = \frac{V_{\gamma,\eta}(\eta^2 X) - V_{\gamma,\eta}(0)}{\eta^6}.$$

Using  $d_\gamma$  from equation (5), define

$$K_\gamma(X) = P_\gamma(X) + \frac{(2048/27)\gamma^4}{|X - d_\gamma e_z|}, \quad (11)$$

where

$$P_\gamma(X) = \gamma H_2(X) + B H_3(X) + \frac{A}{4} H_4(X). \quad (12)$$

**Proposition 4** (First local limit). *Fix  $\gamma > 0$ . For every compact set  $\mathcal{K} \subset \mathbb{R}^3 \setminus \{d_\gamma e_z\}$ , one has*

$$\left\| \tilde{V}_{\gamma,\eta} - (K_\gamma - K_\gamma(0)) \right\| * C^2(\mathcal{K}) = O * \epsilon, \gamma, \mathcal{K}(\eta^2) \quad \text{as } \eta \downarrow 0.$$

*Proof.* Because  $\mathcal{K}$  is compact and does not contain  $d_\gamma e_z$ , all the functions below are defined on a neighborhood of  $\mathcal{K}$  for sufficiently small  $\eta > 0$ . Indeed, the eighth charge remains at  $d_\gamma e_z$ , whereas the other scaled charge positions are  $\eta^{-2}a_j$ ,  $\pm\epsilon\eta^{-2}e_z$ , and  $\pm\eta^{-1}e_z$ , all of which leave every compact set as  $\eta \downarrow 0$ . Define

$$\mathcal{E}_\eta^W(X) = \eta^{-6}\mathcal{E}_\epsilon(\eta^2 X).$$

By homogeneity and the estimate following Proposition 3,

$$\frac{W(\eta^2 X) - W(0)}{\eta^6} = A\eta^{-2}H_2(X) + BH_3(X) + \mathcal{E}^W * \eta(X), \quad \|\mathcal{E}^W * \eta\| * C^2(\mathcal{K}) = O * \epsilon, \mathcal{K}(\eta^2).$$

Now take  $p = p_{\gamma,\eta} = A\eta^3 - \gamma\eta^5$ . Since  $U_{p,\eta}(0) = 2p/\eta$ , equation (9) gives

$$\frac{U_{p,\eta}(\eta^2 X) - U_{p,\eta}(0)}{\eta^6} - (A\eta^{-2} - \gamma)H_2(X) + \left(\frac{A}{4} - \frac{\gamma\eta^2}{4}\right)H_4(X) + \mathcal{E}^{\text{pair}} * \eta(X),$$

where

$$\mathcal{E}^{\text{pair}} * \eta(X) = \eta^{-6}R_{p_{\gamma,\eta},\eta}(\eta^2 X).$$

For sufficiently small  $\eta$ , one has  $|\eta^2 X| \leq \eta/2$  uniformly on  $\mathcal{K}$ . Hence equation (10) and the chain rule imply, for  $|\alpha| \leq 2$ ,

$$\left|D_X^\alpha \mathcal{E}^{\text{pair}} * \eta(X)\right| \leq C|p * \gamma, \eta|\eta^{-1}|X|^{6-|\alpha|}.$$

Since

$$\frac{|p_{\gamma,\eta}|}{\eta} \leq A\eta^2 + \gamma\eta^4,$$

it follows that

$$\|\mathcal{E}^{\text{pair}} * \eta\| * C^2(\mathcal{K}) = O_{\epsilon,\gamma,\mathcal{K}}(\eta^2).$$

The terms  $A\eta^{-2}H_2$  therefore cancel exactly, and the sum of the seed and axial-pair contributions is

$$P_\gamma(X) - \frac{\gamma\eta^2}{4}H_4(X) + \mathcal{E}_\eta^W(X) + \mathcal{E}^{\text{pair}} * \eta(X).$$

Its difference from  $P * \gamma$  is  $O_{\epsilon,\gamma,\mathcal{K}}(\eta^2)$  in  $C^2(\mathcal{K})$ .

Finally, equations (7) and (8) give the exact identity

$$\frac{Q_{\gamma,\eta}}{\eta^6} \left( \frac{1}{|\eta^2 X - \delta e_z|} - \frac{1}{|\delta e_z|} \right) = \frac{2048}{27}\gamma^4 \left( \frac{1}{|X - d_\gamma e_z|} - \frac{1}{d_\gamma} \right).$$

Because  $P_\gamma(0) = 0$ , the right-hand side is exactly the point-charge part of  $K_\gamma(X) - K_\gamma(0)$ . Combining the three contributions proves the stated estimate.  $\square$

## 5 Critical points of the polynomial part

This section and Section 6 both compute Hessians in cylindrical coordinates, so we first record the two elementary facts that will be used in each.

Let  $f$  be smooth near a point with  $\rho > 0$ , and let  $(e_\rho, e_\theta, e_z)$  be the orthonormal cylindrical frame there. The entries of the Hessian bilinear form  $D^2 f$  in this frame are

$$\begin{aligned} D^2 f(e_\rho, e_\rho) &= \partial_{\rho\rho} f, & D^2 f(e_\theta, e_\theta) &= \rho^{-2}\partial_{\theta\theta} f + \rho^{-1}\partial_\rho f, & D^2 f(e_z, e_z) &= \partial_{zz} f, \\ D^2 f(e_\rho, e_z) &= \partial_{\rho z} f, & D^2 f(e_\rho, e_\theta) &= \rho^{-1}\partial_{\rho\theta} f - \rho^{-2}\partial_\theta f, & D^2 f(e_\theta, e_z) &= \rho^{-1}\partial_{\theta z} f. \end{aligned} \quad (13)$$



These identities follow from  $\nabla f = \partial_\rho f e_\rho + \rho^{-1} \partial_\theta f e_\theta + \partial_z f e_z$  together with  $\partial_\theta e_\rho = e_\theta$  and  $\partial_\theta e_\theta = -e_\rho$ . Since the frame is orthonormal, this matrix is the Cartesian Hessian written in a rotated basis; its eigenvalues, determinant, and signature are therefore those of  $D^2 f$ .

**Lemma 5** (Angular decoupling). *Let  $p$  be a critical point of  $f$  with  $\rho > 0$ , and suppose  $\partial_\rho f(p) = \partial_\theta f(p) = 0$ . Then the matrix (13) is block diagonal at  $p$ , with meridional block*

$$D_{\rho,z}^2 f = \begin{pmatrix} \partial_{\rho\rho} f & \partial_{\rho z} f \\ \partial_{\rho z} f & \partial_{zz} f \end{pmatrix}$$

*and a single angular eigenvalue  $\rho^{-2} \partial_{\theta\theta} f$ . In particular  $\det D^2 f = \rho^{-2} \partial_{\theta\theta} f \cdot \det D_{\rho,z}^2 f$ .*

*Proof.* At a critical point  $\partial_\rho f = \partial_\theta f = 0$ , so both correction terms in equation (13) vanish; the hypothesis annihilates the two remaining off-block entries.  $\square$

Every function to which Lemma 5 is applied below has the form  $u(\rho, z) + c\rho^3 \cos(3\theta)$ , with  $c \neq 0$ . For such a function  $\partial_{\theta z} f \equiv 0$  and  $\partial_{\rho\theta} f = -9c\rho^2 \sin(3\theta)$ , which vanishes at every critical point with  $\rho > 0$  because the equation  $\partial_\theta f = -3c\rho^3 \sin(3\theta) = 0$  forces  $\sin(3\theta) = 0$ . The hypothesis of Lemma 5 is therefore automatic.

**Lemma 6** (Saddle dichotomy). *Let  $f$  be harmonic on an open subset of  $\mathbb{R}^3$  and let  $p$  be a nondegenerate critical point of  $f$ . Then the Morse index of  $p$  is 1 or 2; it is 1 when  $\det D^2 f(p) < 0$  and 2 when  $\det D^2 f(p) > 0$ .*

*Proof.* The Hessian is symmetric with vanishing trace and no zero eigenvalue, so its three real eigenvalues are nonzero and sum to zero. Hence at least one is positive and at least one is negative, so the number  $k$  of negative eigenvalues is 1 or 2. The determinant is the product of the eigenvalues, so its sign is  $(-1)^k$ .  $\square$

The polynomial  $P_\gamma$  is a linear combination of  $H_2$ ,  $H_3$ , and  $H_4$ , hence harmonic, so Lemma 6 applies to it.

**Proposition 7** (Critical points of  $P_\gamma$ ). *Let  $\gamma > 0$  and assume*

$$\frac{A\gamma}{B^2} < \frac{3}{80}. \quad (14)$$

*Then  $P_\gamma$  has exactly 21 critical points, all nondegenerate: ten of Morse index 1 and eleven of Morse index 2. As  $\gamma \downarrow 0$ , four lie in an  $O(\gamma)$ -neighborhood of the origin (the origin itself, of index 1, together with three points of index 2 and of norm asymptotic to  $2\gamma/(3B)$ ); eight have norm  $\Theta(\sqrt{\gamma})$ ; and nine converge to distinct nonzero limits. The minimum mutual distance among the eight points of norm  $\Theta(\sqrt{\gamma})$  is  $\Theta(\gamma)$ , even though pairs lying on opposite sides of the plane  $z = 0$  are separated by  $\Theta(\sqrt{\gamma})$ .*

*Proof.* In cylindrical coordinates,

$$P_\gamma = \gamma(\rho^2 - 2z^2) + B\rho^3 \cos(3\theta) + \frac{A}{4}(3\rho^4 - 24\rho^2 z^2 + 8z^4).$$

Its first derivatives are

$$\begin{aligned} \partial_\rho P_\gamma &= \rho(2\gamma + 3B\rho \cos(3\theta) + 3A\rho^2 - 12Az^2), \\ \partial_\theta P_\gamma &= -3B\rho^3 \sin(3\theta), \\ \partial_z P_\gamma &= 4z(-\gamma - 3A\rho^2 + 2Az^2). \end{aligned}$$

When  $\rho = 0$ , the  $x$ - and  $y$ -derivatives vanish identically on the axis, and the remaining Cartesian critical equation is  $4z(-\gamma + 2Az^2) = 0$ , giving the origin and the two axial points listed below. Suppose henceforth that  $\rho > 0$ . Then  $\sin(3\theta) = 0$ , so put  $s = \cos(3\theta) \in \{+1, -1\}$ . If  $z = 0$ , the radial equation is  $2\gamma + 3Bs\rho + 3A\rho^2 = 0$ . It has positive solutions only when  $s = -1$ , in which case it is equation (15). If  $z \neq 0$ , the vertical equation gives

$$z^2 = \frac{\gamma}{2A} + \frac{3}{2}\rho^2.$$

Substitution in the radial equation gives  $15A\rho^2 - 3Bs\rho + 4\gamma = 0$ . This has positive solutions only when  $s = +1$ , and then it is equation (16). Thus the following list is complete.

1. The origin.
2. Two axial points:

$$\rho = 0, \quad z = \pm \sqrt{\frac{\gamma}{2A}}.$$

3. Six planar points. Here  $z = 0$ ,  $\cos(3\theta) = -1$ , and  $\rho$  is either positive root of

$$3A\rho^2 - 3B\rho + 2\gamma = 0. \tag{15}$$

Each radius supplies three angles.

4. Twelve off-planar points. Here  $\cos(3\theta) = 1$ ,  $\rho$  is either positive root of

$$15A\rho^2 - 3B\rho + 4\gamma = 0, \tag{16}$$

and

$$z = \pm \sqrt{\frac{\gamma}{2A} + \frac{3}{2}\rho^2}.$$

Each radius supplies three angles and two signs of  $z$ .

The discriminants of equations (15) and (16) are, respectively,

$$\Delta_p = 9B^2 - 24A\gamma, \quad \Delta_o = 9B^2 - 240A\gamma.$$

Condition (14) is exactly what is needed for  $\Delta_o > 0$ , and it also implies  $\Delta_p > 0$ . The sum and product of the roots in each equation are positive, so both roots of each equation are positive and simple.

We now verify nondegeneracy. At the origin,  $D^2P_\gamma(0) = \text{diag}(2\gamma, 2\gamma, -4\gamma)$ . At either axial point, using  $z^2 = \gamma/(2A)$ , the three Hessian eigenvalues are  $(-4\gamma, -4\gamma, 8\gamma)$ . At a point with  $\rho > 0$  and  $\sin(3\theta) = 0$ , Lemma 5 applies, and the angular eigenvalue is  $\rho^{-2}\partial_{\theta\theta}P_\gamma = -9Bs\rho \neq 0$ . At a planar point,  $s = -1$  and  $z = 0$ . There the mixed entry vanishes,

$$\partial_{\rho z}P_\gamma = \partial_\rho(4z(-\gamma - 3A\rho^2 + 2Az^2)) = -24A\rho z = 0,$$

so the meridional block is itself diagonal, and its two entries, after using equation (15), are the radial and vertical eigenvalues

$$3\rho(2A\rho - B) = \rho \frac{d}{d\rho}(3A\rho^2 - 3B\rho + 2\gamma) \quad \text{and} \quad -4\gamma - 12A\rho^2.$$

The second is negative, and the first is nonzero because the two roots of equation (15) are simple. At an off-planar point, the critical equations and  $s = 1$  give

$$\partial_{\rho\rho}P_\gamma = 3\rho(B + 2A\rho), \quad \partial_{\rho z}P_\gamma = -24A\rho z, \quad \partial_{zz}P_\gamma = 16Az^2.$$

Consequently, the determinant of the meridional  $(\rho, z)$ -block is

$$48A\rho z^2(B - 10A\rho), \quad (17)$$

which is nonzero because

$$\frac{d}{d\rho}(15A\rho^2 - 3B\rho + 4\gamma) = -3(B - 10A\rho)$$

does not vanish at a simple root. The angular eigenvalue is also nonzero. Thus all 21 listed points are nondegenerate.

We next determine the Morse indices. Since  $P_\gamma$  is harmonic, Lemma 6 reduces this to the sign of  $\det D^2P_\gamma$ , which for  $\rho > 0$  factors as in Lemma 5. At the origin the determinant is  $-16\gamma^3 < 0$  and at an axial point it is  $128\gamma^3 > 0$ , so the origin has index 1 and the two axial points index 2. At a planar point the angular eigenvalue  $9B\rho$  is positive and the vertical eigenvalue is negative, so the sign of the determinant is opposite to that of the radial eigenvalue, hence opposite to that of  $\frac{d}{d\rho}(3A\rho^2 - 3B\rho + 2\gamma)$ . This derivative is negative at the smaller root  $\rho_{p,-}$  and positive at the larger root  $\rho_{p,+}$ , so the three points of radius  $\rho_{p,-}$  have index 2 and the three of radius  $\rho_{p,+}$  have index 1. At an off-planar point the angular eigenvalue  $-9B\rho$  is negative, so the sign of the determinant is opposite to that of equation (17), hence opposite to that of  $B - 10A\rho$ . The displayed derivative shows that  $B - 10A\rho > 0$  at the smaller root  $\rho_{o,-}$  of equation (16) and  $B - 10A\rho < 0$  at the larger root  $\rho_{o,+}$ , so the six points of radius  $\rho_{o,-}$  have index 1 and the six of radius  $\rho_{o,+}$  have index 2. In total

$$m_1 = 1 + 3 + 6 = 10, \quad m_2 = 2 + 3 + 6 = 11. \quad (18)$$

We turn to the asymptotics. Recall that  $\rho_{p,-} < \rho_{p,+}$  denote the roots of equation (15) and  $\rho_{o,-} < \rho_{o,+}$  those of equation (16). The quadratic formulas give

$$\rho_{p,\pm} = \frac{3B \pm \sqrt{9B^2 - 24A\gamma}}{6A}, \quad \rho_{o,\pm} = \frac{3B \pm \sqrt{9B^2 - 240A\gamma}}{30A}.$$

Expanding these expressions at  $\gamma = 0$  gives

$$\begin{aligned} \rho_{p,-} &= \frac{2\gamma}{3B} + O(\gamma^2), & \rho_{p,+} &= \frac{B}{A} - \frac{2\gamma}{3B} + O(\gamma^2), \\ \rho_{o,-} &= \frac{4\gamma}{3B} + O(\gamma^2), & \rho_{o,+} &= \frac{B}{5A} - \frac{4\gamma}{3B} + O(\gamma^2). \end{aligned}$$

The origin and the three points with radius  $\rho_{p,-}$  therefore lie in an  $O(\gamma)$ -neighborhood of the origin; the three nonzero points have norm asymptotic to  $2\gamma/(3B)$ . The two axial points and the six off-planar points with radius  $\rho_{o,-}$  have norm  $\Theta(\sqrt{\gamma})$ , because for the latter

$$z^2 = \frac{\gamma}{2A} + O(\gamma^2).$$

The remaining three planar points converge to the three points with

$$\rho = \frac{B}{A}, \quad z = 0, \quad \cos(3\theta) = -1,$$

whereas the six remaining off-planar points converge to the points with

$$\rho = \frac{B}{5A}, \quad z = \pm\sqrt{\frac{3}{2}} \frac{B}{5A}, \quad \cos(3\theta) = 1.$$

These nine limits are nonzero and pairwise distinct. This proves the asserted decomposition  $4 + 8 + 9 = 21$ .

It remains to record the mutual separations of the eight points of norm  $\Theta(\sqrt{\gamma})$ ; these are used in Section 7, and the fact that their minimum separation is *smaller* than their common norm scale is what dictates the radius of the balls employed there. The eight points are the two axial points  $(0, 0, \pm\sqrt{\gamma/(2A)})$  and the six off-planar points of radius  $\rho_{o,-} = 4\gamma/(3B) + O(\gamma^2)$  and heights  $z = \pm\sqrt{\gamma/(2A) + O(\gamma^2)}$ . Thus, although each of the eight has norm  $\Theta(\sqrt{\gamma})$ , the six off-planar ones hug the axis, lying at horizontal distance only  $\rho_{o,-} = \Theta(\gamma)$  from it. The vertical gap between an off-planar point and the axial point with the same sign of  $z$  is

$$\sqrt{\frac{\gamma}{2A} + \frac{3}{2}\rho_{o,-}^2} - \sqrt{\frac{\gamma}{2A}} = O(\gamma^{3/2}),$$

so their distance is  $(\rho_{o,-}^2 + O(\gamma^3))^{1/2} = \rho_{o,-}(1 + O(\gamma)) = \Theta(\gamma)$ . Two off-planar points with the same sign of  $z$  are at distance  $\sqrt{3}\rho_{o,-} = \Theta(\gamma)$ , and every pair with opposite signs of  $z$  has vertical separation  $2\sqrt{\gamma/(2A) + O(\gamma^2)} = \Theta(\sqrt{\gamma})$ . Thus every pair is separated by at least a constant multiple of  $\gamma$ , while the same-sign pairs just described are only a constant multiple of  $\gamma$  apart. The minimum mutual distance among the eight is therefore  $\Theta(\gamma)$ .  $\square$

## 6 Eleven points on the innermost scale

Recall  $d_\gamma$  from equation (5), and write

$$d = d_\gamma = \frac{16}{3}\gamma = \frac{\kappa\gamma}{B}, \quad X = dY.$$

By construction,

$$\frac{Bd}{\gamma} = \kappa = 10, \quad \frac{(2048/27)\gamma^4}{\gamma d^3} = \mu = \frac{1}{2}.$$

By homogeneity, division of equation (11) by  $\gamma d^2$  gives the exact identity

$$\frac{K_\gamma(dY)}{\gamma d^2} = H_2(Y) + 10H_3(Y) + \frac{64A\gamma}{9}H_4(Y) + \frac{1}{2|Y - e_z|}. \quad (19)$$

It therefore converges in  $C^2$  on every compact subset of  $\mathbb{R}^3 \setminus \{e_z\}$  to

$$F(Y) = H_2(Y) + 10H_3(Y) + \frac{1}{2|Y - e_z|}. \quad (20)$$

The function  $F$  is harmonic on  $\mathbb{R}^3 \setminus \{e_z\}$ , being the sum of two harmonic polynomials and a Newtonian kernel, so Lemma 6 applies to it.

**Lemma 8** (Critical points of  $F$ ). *The function  $F$  has exactly 11 critical points, all nondegenerate: four of Morse index 1 and seven of Morse index 2.*

*Proof.* Write  $\mathcal{R} = \sqrt{\rho^2 + (z-1)^2}$ . In cylindrical coordinates the first derivatives of  $F$  are

$$\begin{aligned} \partial_\rho F &= 2\rho + 30\rho^2 \cos(3\theta) - \frac{\rho}{2\mathcal{R}^3}, \\ \partial_\theta F &= -30\rho^3 \sin(3\theta), \\ \partial_z F &= -4z - \frac{z-1}{2\mathcal{R}^3}. \end{aligned}$$

First suppose  $\rho = 0$ . The transverse Cartesian derivatives vanish on the axis. There is no solution with  $z > 1$ , because both terms in  $\partial_z F$  are then negative. For  $z < 1$ , the axial critical equation reduces to

$$z(1-z)^2 = \frac{1}{8}, \quad 0 < z < 1.$$

Here the restriction  $z > 0$  follows from the equation itself. The cubic factors as  $8z(1-z)^2 - 1 = (2z-1)(4z^2-6z+1)$ . Its roots are  $1/2$  and  $(3 \pm \sqrt{5})/4$ ; the root with the plus sign is larger than one. Thus the two admissible roots are

$$z = \frac{1}{2}, \quad z = \frac{3 - \sqrt{5}}{4}. \quad (21)$$

At an axial critical point the Hessian is diagonal. Indeed, in Cartesian coordinates  $F = (x^2 + y^2 - 2z^2) + 10(x^3 - 3xy^2) + \frac{1}{2}\mathcal{R}^{-1}$ ; the Hessian of the cubic term is  $60\begin{pmatrix} x & -y & 0 \\ -y & -x & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , which vanishes at  $x = y = 0$ , while the two remaining terms are invariant under rotation about the  $z$ -axis and hence have Hessians diagonal on that axis. The repeated transverse eigenvalue and the axial eigenvalue are, using  $1/(2(1-z)^2) = 4z$  from the critical equation,

$$\lambda_{\perp} = 2 - \frac{1}{2(1-z)^3} = \frac{2(1-3z)}{1-z}, \quad \lambda_z = -4 + \frac{1}{(1-z)^3} = \frac{4(3z-1)}{1-z}.$$

Neither root in equation (21) equals  $1/3$ , so both axial critical points are nondegenerate. Moreover  $\det D^2F = \lambda_{\perp}^2 \lambda_z$  has the sign of  $3z-1$ ; since  $\frac{1}{2} > \frac{1}{3} > (3-\sqrt{5})/4$ , Lemma 6 gives index 2 at  $z = 1/2$  and index 1 at  $z = (3-\sqrt{5})/4$ .

Now suppose  $\rho > 0$ . The angular equation gives  $s = \cos(3\theta) \in \{+1, -1\}$ . The radial equation is

$$\frac{1}{2\mathcal{R}^3} = 2 + 30s\rho,$$

so admissibility requires  $2 + 30s\rho > 0$ . Substituting this identity into the vertical equation and solving for  $z$  gives

$$z = \frac{2 + 30s\rho}{6 + 30s\rho} \quad (22)$$

and hence  $1-z = 4/(6+30s\rho)$ . Admissibility makes the numerator positive, so  $0 < z < 1$ . The radial equation is therefore equivalent to

$$C_s(\rho) := (2 + 30s\rho) \left( \rho^2 + \frac{16}{(6 + 30s\rho)^2} \right)^{3/2} = \frac{1}{2}. \quad (23)$$

Conversely, every admissible positive root of equation (23), together with any of the three angles having the prescribed value of  $s$ , satisfies all three critical equations.

For  $s = +1$ , put  $h = C_+^2$ . A useful rational form is

$$h(\rho) = \frac{4(15\rho+1)^2(225\rho^4+90\rho^3+9\rho^2+4)^3}{729(5\rho+1)^6}.$$

Direct differentiation gives

$$h'(\rho) = \frac{8\rho(15\rho+1)(225\rho^4+90\rho^3+9\rho^2+4)^2 \Xi(\rho)}{243(5\rho+1)^7}, \quad (24)$$

where

$$\Xi(\rho) = 22500\rho^4 + 14625\rho^3 + 3375\rho^2 + 315\rho - 191.$$

One has

$$\Xi'(\rho) = 90000\rho^3 + 43875\rho^2 + 6750\rho + 315 > 0 \quad (\rho > 0),$$

while  $\Xi(0) = -191$  and  $\Xi(\rho) \rightarrow +\infty$ . Thus the polynomial  $\Xi$  has exactly one positive root, which we call  $\rho_*$ ; it is negative on  $(0, \rho_*)$  and positive on  $(\rho_*, \infty)$ . All other factors in equation (24) are

positive for  $\rho > 0$ . Hence  $h$ , and therefore the positive function  $C_+ = \sqrt{h}$ , is strictly decreasing on  $(0, \rho_*)$  and strictly increasing on  $(\rho_*, \infty)$ . Exact substitution gives

$$C_+(0) > \frac{1}{2}, \quad C_+(1/10) < \frac{1}{2}, \quad C_+(1/4) > \frac{1}{2}. \quad (25)$$

The corresponding squared values are

$$\frac{256}{729}, \quad \frac{4750104241}{21257640000}, \quad \frac{1944701504497}{6347497291776}.$$

The first and third fractions are greater than  $1/4$ , while the middle fraction is less than  $1/4$ ; as  $C_+ > 0$ , the comparisons (25) follow.

We now count the solutions of  $C_+ = 1/2$  on  $(0, \infty)$ . Because  $\rho_*$  is the global minimizer,  $C_+(\rho_*) \leq C_+(1/10) < 1/2$ ; no case distinction according to the position of  $\rho_*$  relative to  $1/10$  is needed. On  $(0, \rho_*]$  the function  $C_+$  decreases strictly from  $C_+(0) = 16/27 > 1/2$  to  $C_+(\rho_*) < 1/2$ , so it attains the value  $1/2$  exactly once there. On  $[\rho_*, \infty)$  it increases strictly from  $C_+(\rho_*) < 1/2$  to  $+\infty$ , since  $C_+(\rho) \sim 30\rho^4$  as  $\rho \rightarrow +\infty$ ; so it attains the value  $1/2$  exactly once there as well. Thus equation (23) has exactly two roots for  $s = +1$ . They are on opposite sides of  $1/10$ , because  $C_+(1/10) < 1/2$  while  $C_+(0) > 1/2$  and  $C_+(1/4) > 1/2$ ; hence one root lies in  $(0, 1/10)$  and the other in  $(1/10, 1/4)$ . Both are simple, since  $C'_+$  vanishes only at  $\rho_*$ , where the value of  $C_+$  is strictly below  $1/2$ .

For  $s = -1$ , admissibility is exactly  $0 < \rho < 1/15$ , and  $C_-(\rho) = C_+(-\rho)$ . On this interval,

$$\Xi(-\rho) < \frac{139}{9} - 191 < 0.$$

Indeed, the positive terms in  $\Xi(-\rho)$  are bounded above by  $22500/15^4 + 3375/15^2 = 139/9$ . In equation (24), evaluated at  $-\rho$ , the factors  $-\rho$  and  $\Xi(-\rho)$  are both negative and every other factor is positive. Thus  $C'_+(-\rho) > 0$ , and consequently  $C'_-(\rho) = -C'_+(-\rho) < 0$ . Moreover,

$$C_-(0) = \frac{16}{27} > \frac{1}{2}, \quad \lim_{\rho \uparrow 1/15} C_-(\rho) = 0.$$

It follows that equation (23) has exactly one simple root for  $s = -1$ .

Each root with  $s = +1$  supplies three angles, and the root with  $s = -1$  supplies three more, giving nine off-axis points. Together with the two points in equation (21), this gives 11 critical points. At an off-axis critical point Lemma 5 applies, and the angular eigenvalue is

$$\rho^{-2} \partial_{\theta\theta} F = -90s\rho \neq 0.$$

We next relate the meridional Hessian determinant directly to the derivative of  $C_s$ . Put

$$a_s(\rho) = 2 + 30s\rho, \quad q(\rho, z) = \frac{1}{2\mathcal{R}^3},$$

and define

$$G_1(\rho, z) = \frac{\partial_\rho F}{\rho} = a_s(\rho) - q(\rho, z), \quad G_2(\rho, z) = \partial_z F = -4z - (z-1)q(\rho, z).$$

At a common zero of  $G_1$  and  $G_2$ , differentiation of  $\partial_\rho F = \rho G_1$  gives

$$\det D_{\rho, z}^2 F = \rho \det D_{(\rho, z)}(G_1, G_2). \quad (26)$$

Now set

$$H = G_2 - (z-1)G_1 = -4z - (z-1)a_s(\rho).$$

The equation  $H = 0$  is exactly equation (22); write its solution as  $z = z_s(\rho)$  and put  $\mathcal{R}_s = \mathcal{R}(\rho, z_s(\rho))$ . Along this graph,

$$g_s(\rho) := G_1(\rho, z_s(\rho)) = \frac{C_s(\rho) - 1/2}{\mathcal{R}_s^3}.$$

At a root of  $C_s = 1/2$ , it follows that  $g'_s = C'_s/\mathcal{R}_s^3$ . Moreover,  $H_z = -(a_s + 4)$ , and implicit differentiation of  $H(\rho, z_s(\rho)) = 0$  gives

$$g'_s = \frac{\det D_{(\rho,z)}(G_1, H)}{H_z} = \frac{\det D_{(\rho,z)}(G_1, G_2)}{H_z};$$

the second equality holds at the common zero because  $H = G_2 - (z - 1)G_1$ . Finally,  $G_1 = 0$  gives  $\mathcal{R}_s^{-3} = 2a_s$ . Substitution in equation (26) therefore yields the transparent identity

$$\det D_{\rho,z}^2 F = -2\rho a_s(\rho)(a_s(\rho) + 4)C'_s(\rho). \quad (27)$$

On the admissible intervals,  $\rho > 0$ ,  $a_s > 0$ , and  $a_s + 4 > 0$ . Since all three roots of equation (23) are simple,  $C'_s(\rho) \neq 0$ ; hence the meridional determinant is nonzero, with sign opposite to that of  $C'_s(\rho)$ . All nine off-axis points are therefore nondegenerate.

Finally we read off the Morse indices from Lemma 6, using the factorization  $\det D^2 F = \rho^{-2} \partial_{\theta\theta} F \cdot \det D_{\rho,z}^2 F$  of Lemma 5. At the root with  $s = +1$  lying in  $(0, \rho_*)$  one has  $C'_+ < 0$ , so the meridional determinant is positive while the angular eigenvalue  $-90\rho$  is negative; the full determinant is negative and the index is 1. At the root with  $s = +1$  lying in  $(\rho_*, \infty)$  one has  $C'_+ > 0$ , so both factors are negative and the index is 2. At the root with  $s = -1$  one has  $C'_- < 0$ , so the meridional determinant is positive while the angular eigenvalue  $+90\rho$  is positive; the index is 2. Each root contributes three points. Adding the two axial points, one of each index, gives

$$m_1 = 1 + 3 = 4, \quad m_2 = 1 + 3 + 3 = 7, \quad (28)$$

which completes the proof.  $\square$

Let  $Y_1, \dots, Y_{11}$  be the critical points in Lemma 8. Choose pairwise disjoint open balls  $U_j$  about them whose compact closures avoid  $e_z$ . By equation (19), the functions

$$Y \mapsto \frac{K_\gamma(dY)}{\gamma d^2}$$

converge to  $F$  in  $C^2$  on  $\bigcup_j \overline{U_j}$ . The implicit-function theorem therefore gives, for every sufficiently small  $\gamma > 0$ , a unique nondegenerate critical point  $Y_j(\gamma)$  near  $Y_j$ , which may be taken in  $U_j$ , with  $Y_j(\gamma) \rightarrow Y_j$ . The invertible change of variables  $X = dY$  and multiplication by the nonzero factor  $\gamma d^2$  preserve critical points and nondegeneracy. Hence

$$X_j(\gamma) = dY_j(\gamma) = dY_j + o(d), \quad j = 1, \dots, 11,$$

are 11 distinct nondegenerate critical points of  $K_\gamma$ .

## 7 The remaining 17 points and the total count

Write

$$S_\gamma(X) = \frac{(2048/27)\gamma^4}{|X - de_z|}, \quad d = d_\gamma = \frac{16}{3}\gamma,$$

so that  $K_\gamma = P_\gamma + S_\gamma$ . For  $j = 0, 1, 2$ , differentiation of the Newtonian kernel gives

$$|D^j S_\gamma(X)| \leq C_j \gamma^4 |X - de_z|^{-1-j}.$$

Call the four critical points of  $P_\gamma$  lying in an  $O(\gamma)$ -neighborhood of the origin the *inner* ones, and the remaining 17 the *outer* ones; the latter comprise the eight of norm  $\Theta(\sqrt{\gamma})$  and the nine with nonzero limits. This section shows that all 17 outer points persist in  $K_\gamma$ .

The persistence of the eight points of norm  $\Theta(\sqrt{\gamma})$  is the delicate case, because their Hessians degenerate at rate  $\gamma$  while their minimum mutual separation is only  $\Theta(\gamma)$ . We therefore record the quantitative form of the implicit function theorem that will be used.

**Lemma 9** (Quantitative implicit function theorem). *Let  $\overline{B}(x_0, r) \subset \mathbb{R}^m$ , and let  $G$  be a  $C^1$  map from an open neighborhood of  $\overline{B}(x_0, r)$  to  $\mathbb{R}^m$ . Let  $L$  be an invertible linear map, put  $\nu := \|L^{-1}\|^{-1} > 0$ , and assume*

$$\|DG(x) - L\| \leq \frac{\nu}{2} \quad \text{for all } x \in \overline{B}(x_0, r), \quad \text{and} \quad |G(x_0)| \leq \frac{\nu r}{2}.$$

*Then  $G$  has exactly one zero  $x_*$  in  $\overline{B}(x_0, r)$ , it satisfies  $|x_* - x_0| \leq 2|G(x_0)|/\nu$ , and  $DG(x_*)$  is invertible.*

*Proof.* Put  $\Phi(x) = x - L^{-1}G(x)$ , so that  $D\Phi(x) = L^{-1}(L - DG(x))$  and hence  $\|D\Phi\| \leq \nu^{-1} \cdot \nu/2 = 1/2$  on the ball. Thus  $\Phi$  is a  $\frac{1}{2}$ -contraction there. Moreover  $|\Phi(x_0) - x_0| = |L^{-1}G(x_0)| \leq |G(x_0)|/\nu \leq r/2$ , so for  $x \in \overline{B}(x_0, r)$ ,

$$|\Phi(x) - x_0| \leq |\Phi(x) - \Phi(x_0)| + |\Phi(x_0) - x_0| \leq \frac{1}{2}|x - x_0| + \frac{r}{2} \leq r.$$

Hence  $\Phi$  maps the ball into itself and the Banach fixed point theorem supplies a unique fixed point  $x_*$ ; as  $L$  is invertible, the fixed points of  $\Phi$  are exactly the zeros of  $G$ . The contraction estimate gives  $|x_* - x_0| \leq (1 - \frac{1}{2})^{-1}|\Phi(x_0) - x_0| \leq 2|G(x_0)|/\nu$ . Finally,

$$\|L^{-1}(DG(x_*) - L)\| \leq \frac{1}{2},$$

so  $DG(x_*) = L(I + L^{-1}(DG(x_*) - L))$  is invertible by the Neumann-series criterion.  $\square$

Consider now the eight critical points of  $P_\gamma$  on the  $O(\sqrt{\gamma})$  scale. At the two axial points, the eigenvalues divided by  $\gamma$  are  $(-4, -4, 8)$ . At each of the six off-planar points with radius  $\rho_{o,-}$ , the angular eigenvalue and the meridional Hessian, again divided by  $\gamma$ , have the limits

$$\gamma^{-1}\lambda_\theta \longrightarrow -12, \quad \gamma^{-1}D_{\rho,z}^2 P_\gamma \longrightarrow \begin{pmatrix} 4 & 0 \\ 0 & 8 \end{pmatrix}.$$

These limits follow from  $\rho_{o,-} = 4\gamma/(3B) + O(\gamma^2)$  and  $z^2 = \gamma/(2A) + O(\gamma^2)$ . Since an orthonormal cylindrical frame is an orthogonal change of basis, the eight Cartesian Hessians divided by  $\gamma$  therefore have all eigenvalues bounded away from zero, uniformly for small  $\gamma$ . Thus there are  $\gamma_0 > 0$  and  $\nu_0 > 0$  such that  $\|(\gamma^{-1}D^2 P_\gamma(X_0))^{-1}\|^{-1} \geq \nu_0$  at each of the eight points  $X_0$  whenever  $0 < \gamma < \gamma_0$ .

By the last paragraph of the proof of Proposition 7, the minimum mutual distance among these eight points is  $\Theta(\gamma)$ , because the six off-planar points lie at horizontal distance only  $\rho_{o,-} = \Theta(\gamma)$  from the axis. Consequently they admit mutually disjoint closed balls of radius  $c\gamma$  once  $c > 0$  is a fixed sufficiently small constant. On such a ball,

$$\|D^2 P_\gamma(X) - D^2 P_\gamma(X_0)\| \leq c\gamma \sup_{\overline{B}(X_0, c\gamma)} |D^3 P_\gamma| = O(c\gamma),$$



because  $D^3P_\gamma = B D^3H_3 + (A/4)D^3H_4$ , in which  $D^3H_3$  is a constant tensor and  $D^3H_4$  is linear in  $X$ ; on the balls this is bounded by an absolute multiple of  $B + A\sqrt{\gamma} = O(1)$ . Shrinking  $c$  therefore makes  $\|D^2P_\gamma(X) - D^2P_\gamma(X_0)\| \leq \nu_0\gamma/4$  throughout the ball.

On these balls,  $|X - de_z| \asymp \sqrt{\gamma}$ , since  $d = \Theta(\gamma)$ . Consequently,

$$|\nabla S_\gamma| = O(\gamma^3), \quad |D^2S_\gamma| = O(\gamma^{5/2}) = o(\gamma).$$

We apply Lemma 9 with  $G = \nabla K_\gamma$ ,  $m = 3$ ,  $r = c\gamma$ , and  $L = D^2P_\gamma(X_0)$ , so that  $\nu \geq \nu_0\gamma$ . The first hypothesis holds because

$$\|D^2K_\gamma(X) - L\| \leq \|D^2P_\gamma(X) - D^2P_\gamma(X_0)\| + \|D^2S_\gamma(X)\| \leq \frac{\nu_0\gamma}{4} + O(\gamma^{5/2}) \leq \frac{\nu_0\gamma}{2} \leq \frac{\nu}{2}$$

for all small  $\gamma$ . The second holds because  $\nabla P_\gamma(X_0) = 0$  gives  $|G(X_0)| = |\nabla S_\gamma(X_0)| = O(\gamma^3)$ , which is at most  $\nu_0\gamma \cdot c\gamma/2 \leq \nu r/2$  for small  $\gamma$ . The lemma therefore yields exactly one critical point of  $K_\gamma$  in each ball, it is nondegenerate, and its displacement from  $X_0$  is  $O(\gamma^{-1}\gamma^3) = O(\gamma^2) = o(\gamma)$ . As the balls are disjoint, the eight continuations are distinct. Moreover, throughout each ball the straight-line family from  $D^2P_\gamma(X_0)$  to  $D^2K_\gamma(X)$  remains invertible by the same norm estimate. Thus each continuation has the same Morse index as its corresponding critical point of  $P_\gamma$ .

The remaining nine outer critical points of  $P_\gamma$  converge to distinct nonzero limits. At a limiting planar point, the angular, radial, and vertical eigenvalues are, respectively,

$$\frac{9B^2}{A}, \quad \frac{3B^2}{A}, \quad -\frac{12B^2}{A},$$

and hence are nonzero. At a limiting off-planar point, equation (17) has the nonzero limit  $-72B^4/(125A^2)$ , while the angular eigenvalue tends to  $-9B^2/(5A)$ . Thus all nine limiting Hessians are invertible.

Put  $P_0 = BH_3 + (A/4)H_4$ . Choose fixed pairwise disjoint neighborhoods of the nine limiting critical points of  $P_0$  whose closures avoid the origin. On their union,  $P_\gamma \rightarrow P_0$  and  $\|S_\gamma\| * C^2 = O(\gamma^4)$ , the latter because  $de_z \rightarrow 0$ . Consequently  $K * \gamma = P_\gamma + S_\gamma \rightarrow P_0$  in  $C^2$  there. The implicit-function theorem therefore gives nine distinct nondegenerate critical points of  $K_\gamma$ , one near each limit. Their Hessians converge to the corresponding invertible limiting Hessians, so their Morse indices are unchanged. We have proved that all 17 outer critical points of  $P_\gamma$  persist in  $K_\gamma$ , with unchanged Morse indices.

The 11 inner critical points found in Section 6 satisfy  $|X| = O(d) = O(\gamma)$ , whereas the preceding 17 points have distance  $\asymp \sqrt{\gamma}$  or  $\asymp 1$  from the origin. Thus the two families are disjoint for sufficiently small  $\gamma$ .

Thus  $K_\gamma$  has at least

$$17 + 11 = 28 \tag{29}$$

nondegenerate critical points for sufficiently small  $\gamma > 0$ . Choose  $\gamma$  small enough that this conclusion and all preceding smallness conditions hold, and fix it. Choose pairwise disjoint open balls about the 28 points whose compact closures avoid  $de_z$ . The  $C^2$ -convergence in Proposition 4 and the implicit-function theorem give one nondegenerate critical point of  $\tilde{V}_{\gamma,\eta}$  in each ball for all sufficiently small  $\eta > 0$ . The invertible rescaling  $x = \eta^2 X$ , subtraction of a constant, and multiplication by  $\eta^{-6}$  preserve critical points and nondegeneracy, so these yield 28 distinct critical points of  $V_{\gamma,\eta}$ . Since their  $X$ -coordinates remain bounded, all 28 lie at distance  $O(\eta^2)$  from the origin in the original variables.

The 23 specified nonzero critical points of  $W$  have positive distance from the origin. Choose mutually disjoint open balls about them whose compact closures avoid the five original charge positions and the origin. On the union of these closures, the potentials of the second axial

pair and the eighth charge tend to zero in  $C^2$ , because their strengths are respectively  $O(\eta^3)$  and  $O(\eta^8)$ , while their positions tend to the origin. All 23 critical points therefore persist as nondegenerate critical points of  $V_{\gamma,\eta}$ . They remain a fixed positive distance from the origin and are consequently disjoint from the 28 local points.

Finally, choose  $\eta > 0$  small enough that both persistence conclusions hold and the explicit positivity and distinctness conditions in Section 3 are satisfied. This respects the parameter order  $\epsilon$ , then  $\gamma$ , then  $\eta$ , and produces an eight-charge configuration for which  $V_{\gamma,\eta}$  has at least

$$23 + 28 = 51. \quad (30)$$

For comparison, if the eighth charge is omitted, all 21 critical points of  $P_\gamma$  persist and the same argument gives  $23 + 21 = 44$  specified critical points for seven charges.

*Remark 10 (Bookkeeping).* It is suggestive to describe the eighth charge as replacing the four inner  $O(\gamma)$ -scale points of  $P_\gamma$  by 11 points, so that  $44 - 4 + 11 = 51$ . This is only a mnemonic. The proof instead supplies  $23 + 17 + 11 = 51$  distinct nondegenerate critical points. It neither asserts nor requires the four inner points of  $P_\gamma$  to disappear; any such persistence would only increase the count.

## 8 Removing all remaining degeneracies

The 51 counted points are already nondegenerate and therefore isolated, but the preceding argument does not exclude additional degenerate critical points. We now remove that possibility while preserving the 51 points. Keep the eight positions fixed and label them  $b_1, \dots, b_8$ , with  $b_1, b_2, b_3$  the three triangle vertices. Let  $q^{(0)}$  be the strength vector constructed above, and vary  $q = (q_1, \dots, q_8)$  in the open positive orthant  $\mathcal{P} = (0, \infty)^8$ .

On  $M = \mathbb{R}^3 \setminus \{b_1, \dots, b_8\}$ , consider the smooth family of electric fields

$$E(q, x) = \sum_{i=1}^8 q_i \frac{x - b_i}{|x - b_i|^3} = -\nabla V_q(x).$$

For fixed  $(q, x) \in \mathcal{P} \times M$ , the derivatives with respect to the strengths are

$$\partial_{q_i} E(q, x) = \frac{x - b_i}{|x - b_i|^3}.$$

These vectors span  $\mathbb{R}^3$ . Indeed, if a nonzero vector  $v$  were orthogonal to all of them, then  $v \cdot b_i = v \cdot x$  for every  $i$ , so all eight positions would lie in one affine plane. This is impossible: the three noncollinear points  $b_1, b_2, b_3$  determine the plane  $z = 0$ , whereas every axial charge has nonzero  $z$ -coordinate. Thus the partial derivative  $D_q E(q, x)$  is surjective at every  $(q, x)$ , and hence  $E : \mathcal{P} \times M \rightarrow \mathbb{R}^3$  is a submersion.

Choose pairwise disjoint neighborhoods in  $M$  of the 51 constructed critical points. Applied at these points, the implicit-function theorem and continuity of the Hessian give an open neighborhood  $\mathcal{U} \subset \mathcal{P}$  of  $q^{(0)}$  such that every  $q \in \mathcal{U}$  has a distinct nondegenerate critical point in each of the 51 neighborhoods. Since a submersion is transverse to  $\{0\}$ , the parametric transversality theorem [9, p. 68] implies that, for almost every  $q \in \mathcal{P}$ , the partial map  $E(q, \cdot)$  is transverse to  $\{0\}$ . Such parameters are dense in  $\mathcal{P}$ , so we may choose  $q' \in \mathcal{U}$  with this property. At a zero of  $E(q', \cdot)$ ,  $D_x E(q', x) = -D^2 V_{q'}(x)$ . Consequently, transversality to  $\{0\}$  says exactly that every critical point of  $V_{q'}$  is nondegenerate, while the choice  $q' \in \mathcal{U}$  preserves the 51 constructed points.

Every critical point lies in the convex hull of the charge positions. At a zero of  $E$ , one has

$$x = \frac{\sum_i (q'_i/|x - b_i|^3) b_i}{\sum_i q'_i/|x - b_i|^3},$$

which is a convex combination because all  $q'_i > 0$ . Near  $b_i$ , the term  $q'_i(x - b_i)/|x - b_i|^3$  has norm  $q'_i/|x - b_i|^2$ , while the other seven terms remain bounded in a fixed ball about  $b_i$  containing no other charge. Positivity of  $q'_i$  therefore permits a radius  $r_i > 0$  such that

$$\frac{q'_i}{|x - b_i|^2} \gg \left| \sum_{j \neq i} q'_j * j \frac{x - b_j}{|x - b_j|^3} \right| \quad (0 < |x - b_i| < r_i).$$

Thus  $E(q', x) \neq 0$  in each punctured ball  $B * r_i(b_i) \setminus \{b_i\}$ . Shrinking the radii if necessary, assume that these balls are pairwise disjoint.

Let  $H$  be the convex hull of the charge positions and set

$$K = H \setminus \bigcup_{i=1}^8 B_{r_i}(b_i).$$

The preceding two paragraphs show that every critical point belongs to the compact subset  $K \subset M$ . Because  $E(q', \cdot)$  is continuous on  $K$ , its zero set there is closed and hence compact. Each zero is isolated because its derivative is invertible. An infinite subset of a compact metric space has an accumulation point, so this compact set of isolated zeros must be finite. The potential  $V_{q'}$  is therefore Morse and has a finite number of critical points, at least 51. This completes the proof of Theorem 1.

## 9 The Morse-index ledger

Every construction above produces critical points together with their Morse indices, and these indices satisfy a global constraint. We record it, both as an independent check on the count of Theorem 1 and because it localizes what any further critical points would have to look like.

**Proposition 11** (Index identity). *Let  $V = \sum_{i=1}^n q_i |x - b_i|^{-1}$  with all  $q_i > 0$  and all  $b_i$  distinct, and suppose  $V$  is Morse. Let  $m_1$  and  $m_2$  denote the numbers of critical points of Morse index 1 and 2. Then every critical point has index 1 or 2, both numbers are finite, and*

$$m_2 - m_1 = n - 1. \quad (31)$$

*Proof.* The compactness argument in Section 8 applies verbatim to any finite set of distinct positive charges: every critical point lies in the convex hull of the charge positions, and a neighborhood of each charge is free of critical points. Since  $V$  is Morse, its critical set is therefore compact and discrete, hence finite. Moreover,  $V$  is harmonic off the charges, so Lemma 6 shows that every critical point has index 1 or 2.

Put  $E = -\nabla V$ . Choose  $R$  so large that  $\overline{B}_R(0)$  contains all charges and all critical points in its interior, and choose  $r > 0$  so small that the closed balls  $\overline{B}_r(b_i)$  are disjoint, contained in  $B_R(0)$ , and free of critical points. On the compact region  $\Omega = \overline{B}_R(0) \setminus \bigcup_i \overline{B}_r(b_i)$  the field  $E$  has no zeros on the boundary, so the additivity and boundary formula for Brouwer degree give

$$\sum_{p: E(p)=0} \text{ind}(E, p) = \deg \left( \frac{E}{|E|} \Big|_* \partial B_R(0) \right) - \sum *i = 1^n \deg \left( \frac{E}{|E|} \Big|_{\partial B_r(b_i)} \right),$$

the inner boundaries entering with the opposite sign. The mean-value theorem applied to  $u \mapsto u|u|^{-3}$  gives

$$E(x) = Q \frac{x}{|x|^3} + O(|x|^{-3}), \quad Q := \sum_i q_i > 0,$$

uniformly as  $|x| \rightarrow \infty$ . On  $\partial B_R(0)$ , the leading term has norm  $Q/R^2$ , whereas the error is  $O(R^{-3})$ . For all sufficiently large  $R$ , the straight-line homotopy from  $E$  to the outward radial field  $Qx/|x|^3$  is therefore nowhere zero, and the outer degree is 1. Likewise, near  $b_i$ ,

$$E(x) = q_i \frac{x - b_i}{|x - b_i|^3} + O(1).$$

On  $\partial B_r(b_i)$ , the first term has norm  $q_i/r^2$ ; hence, for all sufficiently small  $r$ , the analogous straight-line homotopy is nowhere zero and the degree with the standard outward orientation of  $\partial B_r(b_i)$  is 1. The boundary orientation inherited from  $\Omega$  is the opposite orientation on every inner sphere, which explains the minus signs above. Hence the index sum equals  $1 - n$ .

At a nondegenerate zero  $p$  of  $E$  of Morse index  $k$ ,

$$\text{ind}(E, p) = \text{sign det } DE(p) = \text{sign det } (-D^2V(p)) = (-1)^3(-1)^k = (-1)^{k+1}.$$

Therefore  $\sum_p (-1)^{k(p)+1} = 1 - n$ , that is,  $\sum_p (-1)^{k(p)} = n - 1$ . Since  $k \in \{1, 2\}$ , the left-hand side is  $m_2 - m_1$ .  $\square$

For completeness, consider first two positive charges separated by distance  $L$ . The convex-combination formula in Section 8 places every critical point at  $b_1 + te$ , where  $e = (b_2 - b_1)/L$  and  $0 < t < L$ . The field equation reduces to

$$\frac{q_1}{t^2} = \frac{q_2}{(L - t)^2},$$

which has the unique solution  $t = L\sqrt{q_1}/(\sqrt{q_1} + \sqrt{q_2})$ . At this point the Hessian is

$$D^2V = S(3ee^\top - I), \quad S = \frac{q_1}{t^3} + \frac{q_2}{(L - t)^3} > 0.$$

Its eigenvalues are  $2S, -S, -S$ , so the unique critical point is nondegenerate of index 2, consistently with Proposition 11.

We also verify directly the indices of the three noncentral triangle points used below. By symmetry, it suffices to work on the negative  $x$ -axis and write the point as  $(-t, 0, 0)$ , with  $0 < t < 1/2$ . The  $x$ -component of the electric field there is

$$F(t) = -\frac{1}{(1 + t)^2} + \frac{1 - 2t}{(t^2 - t + 1)^{3/2}}.$$

Set

$$h(t) = \log \left( \frac{(1 - 2t)(1 + t)^2}{(t^2 - t + 1)^{3/2}} \right).$$

Then  $F(t) = (1 + t)^{-2}(e^{h(t)} - 1)$ , while

$$h'(t) = \frac{3(1 - 7t + 4t^2)}{2(1 - 2t)(1 + t)(t^2 - t + 1)}.$$

The numerator of  $h'$  has exactly one zero in  $(0, 1/2)$ , namely  $(7 - \sqrt{33})/8$ . Thus  $h$  first increases and then strictly decreases; because  $h(0) = 0$  and  $h(t) \rightarrow -\infty$  as  $t \uparrow 1/2$ , it has a unique zero  $t_*$  in

that interval, and  $F'(t_*) < 0$ . Reflection symmetry makes the Hessian of  $V_\Delta$  diagonal at  $(-t_*, 0, 0)$ . Since  $E = -\nabla V_\Delta$  and the coordinate on the ray is  $x = -t$ , one has  $\partial_{xx}V_\Delta = F'(t_*) < 0$ . Also

$$\partial_{zz}V_\Delta = -\sum_{i=1}^3 |(-t_*, 0, 0) - a_i|^{-3} < 0.$$

Harmonicity gives  $\partial_{yy}V_\Delta = -(\partial_{xx}V_\Delta + \partial_{zz}V_\Delta) > 0$ . Hence this point is nondegenerate of index 2, as are its two rotations. At the center, the expansion  $V_\Delta = 3 + \frac{3}{4}H_2 + \dots$  gives the Hessian  $\frac{3}{4}\text{diag}(2, 2, -4)$ , so the center has index 1.

We now assemble the ledger for the configuration of Theorem 1. For this we take the 23 seed points to be those produced by the construction of [8, Lemmas 1–2 and Remark 2]: the three noncentral critical points of the triangle together with the 20 nonzero members of the 21-point family on the scale  $\epsilon^2$ . The proof of Proposition 7 uses only that the three coefficients are positive and satisfy the smallness condition (14), so the proposition applies verbatim to any polynomial  $\gamma H_2 + B H_3 + \frac{A}{4} H_4$  with those properties. The 21-point family of [8] is the critical set of  $\frac{5}{32}H_2 + \frac{15}{8}H_3 + \frac{3}{16}H_4$ , that is, of the triple

$$A = \frac{3}{4}, \quad B = \frac{15}{8}, \quad \gamma = \frac{5}{32}, \quad \text{for which} \quad \frac{A\gamma}{B^2} = \frac{1}{30} < \frac{3}{80}.$$

Equation (18) therefore applies to it: it has  $m_1 = 10$  and  $m_2 = 11$ . Discarding the origin, which has index 1, and adjoining the three noncentral triangle points, which have index 2, the 23 seed points have  $m_1 = 9$  and  $m_2 = 14$ . For the 17 outer points we discard from equation (18) the origin (index 1) and the three inner planar points of radius  $\rho_{p,-}$  (index 2), leaving  $m_1 = 9$  and  $m_2 = 8$ . The 11 innermost points have  $m_1 = 4$  and  $m_2 = 7$  by equation (28). Morse indices are unchanged by all the continuations used above: along each of them the Hessian varies continuously and remains invertible, so no eigenvalue can cross zero and the signature is constant. This includes the final perturbation from  $q^{(0)}$  to  $q'$  in Section 8: the common implicit-function neighborhood may be taken to be a ball about  $q^{(0)}$ , and  $q'$  may be chosen inside that ball. Every rescaling used in the construction is of the form  $g(X) = cf(sX) + C$  with  $c, s > 0$ , and hence  $D^2g(X) = cs^2D^2f(sX)$ ; it also preserves the signature. Hence:

| Family                        | $m_1$ | $m_2$ | total |
|-------------------------------|-------|-------|-------|
| 23 seed points (Theorem 2)    | 9     | 14    | 23    |
| 17 outer points (Section 7)   | 9     | 8     | 17    |
| 11 innermost points (Lemma 8) | 4     | 7     | 11    |
| Total                         | 22    | 29    | 51    |

The ledger closes:  $29 - 22 = 7 = 8 - 1$ , in exact agreement with equation (31) for  $n = 8$ . The same computation for the specified points in the seven-charge construction of Section 7 gives  $m_1 = 9 + 10 = 19$  and  $m_2 = 14 + 11 = 25$ , and  $25 - 19 = 6 = 7 - 1$ ; and for the 24 specified points of the five-charge seed it gives  $m_1 = 10$ ,  $m_2 = 14$ , and  $14 - 10 = 4 = 5 - 1$ .

Two consequences deserve mention. First, the index computation provides a global consistency check on the assembled family counts: the independently computed Hessian signatures satisfy the topological identity exactly. Second, the 51 constructed points already contribute the full difference  $m_2 - m_1 = 7$ . Proposition 11 therefore implies that the numbers of any additional index-1 and index-2 critical points of  $V_{q'}$  must be equal. In particular, any additional points occur in an even total number. The construction is consistent with  $N = 51$  exactly, although it does not prove that no further critical points exist.

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